

Third Terminal Examination - 2079

Class : XI

Subject: Chemistry (3011)

Full Marks: 75

Time: 3 hrs.

SET 'A'

Pass Marks: 30

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

Group 'A'

Rewrite the best alternative of each question in your answer sheet. [11 × 1 = 11]

1. Which one of the following pairs of compounds illustrates the law of multiple proportions?



2. Which of the following set of quantum number is not possible for 4p electron?

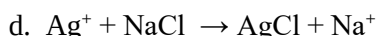
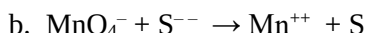
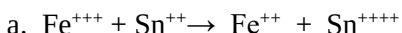
a. $n=4, l=1, m=-1, s=+\frac{1}{2}$

b. $n=4, l=1, m=0, s=+\frac{1}{2}$

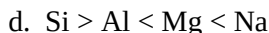
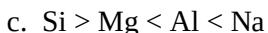
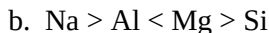
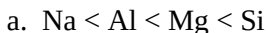
c. $n=4, l=1, m=2, s=+\frac{1}{2}$

d. $n=4, l=1, m=-1, s=-\frac{1}{2}$

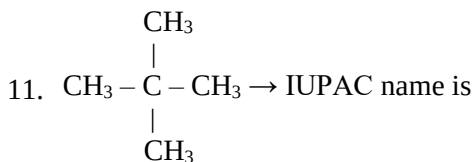
3. Which of the following is not a redox reaction?



4. Correct order of increasing I.E. is



5. For the precipitation of Group IIIB metal ions as their sulphides, H_2S gas is passed in presence of
- dil.HCl
 - NH_4OH
 - $\text{NH}_4\text{OH} + \text{NH}_4\text{Cl}$
 - $\text{NH}_4\text{OH} + (\text{NH}_4)_2\text{CO}_3$
6. Ammonia can be dried by using
- Conc. Sulphuric acid
 - Quick lime
 - Phosphorous pentoxide
 - Fused calcium chloride
7. Fluorine shows only one oxidation state because
- Its atom is smaller in size
 - It is volatile
 - It is highly electronegative
 - It has no d-orbital
8. Which one of the following is not the isomer of the other three?
- pentane
 - 2-methylbutane
 - 2,2-dimethylpropane
 - 2,3-dimethylbutane
9. A fuel having octane number 60 produces the same extent of knocking as produced by fuel mixture containing
- 60% isooctane and 40% n-methylnaphthalene
 - 60% n-heptane and 40% n-hexadecane
 - 40% n-methylnaphthalene and 60% n-heptane
 - 60% isooctane and 40% n-heptane
10. Which of the following is functional group of ester?
- $-\text{COOH}$
 - $-\text{COOR}$
 - $-(\text{CO})_2\text{O}$
 - $-\text{CONH}_2$



- Dimethyl hexane
- Tetra methane
- Neo pentane
- 2,2-Dimethyl propane

Group - B

Give short answer to the following questions.

[8 × 5 = 40]

1. Rutherford's nuclear model of atoms is based on alpha particle scattering experiments. During alpha particle scattering experiments, alpha particles emitted from a radioactive source are bombarded on thin gold foil and the fate of these particles was observed on a movable zinc sulphide screen.
 - a. What are alpha particles? [1]
 - b. Which observation led Rutherford to conclude that
 - i. Most of the mass of an atom is concentrated in the nucleus.
 - ii. Atom is almost empty. [2]
 - c. Draw the figure showing an atom's deflection of alpha particles. What are the limitations of this model? [2]
2. a. The reaction of sulphur with sodium hydroxide to give sodium sulphide and sodium thiosulphate is a redox reaction.
$$\text{S} + \text{NaOH} \longrightarrow \text{Na}_2\text{S} + \text{Na}_2\text{S}_2\text{O}_3 + \text{H}_2\text{O}$$
Balance the redox reaction with help of oxidation number method.
 - b. Calculate the oxidation number of the underlined elements.
 - i) (COOH)₂
 - b. KMnO₄ [3+2]
3. a. Whenever substances are made to react, they follow certain laws to give the products. These laws are called laws of chemical combination.
 - i. State the law of constant proportion. [1]
 - ii. The formation of CO and CO₂ illustrate the law of multiple proportions. Justify. [2]
- b. 34.2 gram of sucrose (C₁₂H₂₂O₁₁) are dissolved in 18 grams of water. Calculate the number of oxygen atoms in the solution. [2]

OR

Electronegativity value of IA group elements and third-period elements are given below in Pauling's scale.

Atom (Group I)	Electronegativity Value	Atom (Group 17)	Electronegativity
Li	1.0	F	4.0
Na	0.9	Cl	3.0
K	0.8	Br	2.8
Rb	0.8	I	2.5
Ca	0.7	At	2.2

- i. Define the term 'Electronegativity'. How does this differ from electron affinity? [2]
 - ii. Why does electronegativity decrease down the group? [1]
 - iii. What are the advantages of the modern periodic table? [2]
4. Kipp's apparatus, also called Kipp generator, is an apparatus designed for preparation of small volumes of gases. It was invented around 1844 by the Dutch pharmacist Petrus Jacobus Kipp and widely used in chemical laboratories. In Kipp's apparatus H_2S is prepared by reacting FeS with dil. H_2SO_4 .
- i. What happens when H_2S is passed through the acidified KMnO_4 ? [1]
 - ii. How H_2S gas is used in the qualitative salt analysis? [1]
 - iii. Write the theory involved in manufacture of sulphuric acid from contact process. Also, mention condition for maximum yield. [3]

5. Oxygen is the third most abundant element by mass which readily forms oxides with other elements. Some of the oxides are given below

Na_2O	Al_2O_3	CO	SO_2	Fe_3O_4	H_2O_2
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- i. Identify the acidic oxide, basic oxide, neutral oxide and mixed oxide from the above table. [2]
 - ii. What happens when CO is heated with
 - a. Fehling's solution
 - b. Ni metal[2]
 - iii. Why is CO a toxic gas? [1]
6. Chlorine is a good bleaching agent. It is used for bleaching textiles, paper, wood, pulp, etc.
- i. How does chlorine bleach the colored organic matter? [2]
 - ii. Can iodine be used instead of chlorine for bleaching purposes? [1]
 - iii. Write the balanced chemical reaction for the preparation of chlorine with the application of heat and without the application of heat. [2]

OR

Nitrogen is most important constituent of protein; hence all plants and animals contain nitrogen.

- a. Discuss the principle involved in the manufacture of ammonia by Haber's process. Also, draw the flow sheet diagram. [1.5+1.5]
- b. Write the action of ammonia on:
 - i) mercurous nitrate
 - ii) FeCl_3[2]

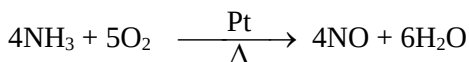
7. Isomerism is the phenomenon in which more than one compounds have the same chemical formula but different chemical structures.
- What is functional isomerism? Write the functional isomers of formula C_3H_6O with structure and IUPAC names. [1+2]
 - Write the IUPAC name of the following organic name: [2]
 - $CH_3 - CH_2 - CH_2 - Cl$
 - $$\begin{array}{c}
 CH_3 \\
 | \\
 CH_3 - CH_2 - C - OH \\
 | \\
 Br
 \end{array}$$
8. Organic chemistry is the study of the structure, properties, composition, reaction and preparation of carbon-containing compounds.
- What is homologous series? Write the characteristics of homologous series. [1+2]
 - Define electrophile and nucleophile with an example of each. [2]

Group - C

Give long answer to the following questions.

[3 × 8 = 24]

9. a. In the given chemical equation,



- Determine the limiting reagent and calculate the mass of nitric oxide produced by the reaction of 2 moles of ammonia with 2 moles of oxygen. [2]
- Calculate the volume of HCl gas required at NTP which can completely react with 2 moles of ammonia gas to produce ammonium chloride [2]

- b. Write the postulates of the Bohr atomic model. Write any two drawbacks of it. [3+1]

OR

Chemistry is a branch of science that deals with the composition, structure, and properties of substances and the change they undergo.

- a. State Faraday's first law of electrolysis. Calculate the thickness of silver deposited on the spoon of area 800 cm^2 when 0.2 ampere current is passed through the solution for 3 hours. (Density of silver = 10.5 g/cc , atomic weight = 108) [1+3]
- b. What are flux and slag? Give examples. [2]
- c. What is tautomerism? Discuss with example. [2]
10. Hydrogen gas is a direct source of clean energy and has applications in food industries, petrochemical refineries, cutting/welding, metallurgy, etc.
- i. What is nascent hydrogen?
- ii. Mention two examples to show nascent hydrogen is a more powerful reducing agent than molecular hydrogen.
- iii. Write the differences between nascent hydrogen and molecular hydrogen.
- iv. Write the important applications of hydrogen peroxide.
- v. Write the names of two metals that can displace hydrogen gas from dilute nitric acid with a chemical reaction. [1+2+2+1+2]
11. a. Lassaigne's test can be performed for the detection of hetero-elements present in organic compounds by fusing the compound with active metal like sodium.

- i. What is the purpose of preparing sodium extract or Lassaigne's solution? [1]
- ii. Write the reactions involved in the detection of Nitrogen in the organic compound? [2]
- iii. Why sodium extract is alkaline in nature? [1]
- b. Write the structure of following IUPAC names: [2]
 - i. 2,3-dimethyl-but-2-ene
 - ii. Pent-4-en-2-ol
- c. What is a chemical bond? Define the Octet rule. [2]

❧ - The End - ❧